

Problem 8.1 (MATLAB)

Principal Components Analysis (PCA)

The file '**dataset_problem_8_1.mat**' contains data for male national track records for 54 countries in the following 8 running disciplines (time units indicated): 100m [sec], 200m [sec], 400m [sec], 800m [min], 1500m [min], 5000m [min], 10000m [min] and Marathon [min].

The ID of the 54 countries can be found in the file '**countriesID_problem_8_1.dat**'.

It is desired to simplify the interpretation of the 8-dimensional dataset by performing a Principal Component Analysis. Since the 8 variables have different units and different amounts of variation, it is most appropriate to perform the PCA on the sample correlation matrix, R, instead of on the sample covariance matrix, S, directly.

PCA on R (for time-data):

- Load the dataset file
- Calculate descriptive statistics S and R
- Make a boxplot of the male time-data
- Make a scatter plot (matrix scatter plot) of the male time-data

- Perform a PCA on R using ***svd*** or ***eig***
- Calculate the fraction of the total variation explained by each of the Principal Components (PC's) and visualize this in a Pareto plot and a scree plot

- Indicate the loadings for PC1 (the PC with highest variance explained/highest eigenvalue) on Z-variables (standardized variables) and X-variables respectively
- Interpret the loadings for PC1 (easiest for Z-loadings)
- Calculate the PC1 scores for the 54 countries
- Using ***sort***, find the 10 "best" countries with respect to PC1(lowest scores)

- Indicate the loadings for PC2 (the PC with second highest variance explained/second highest eigenvalue) on Z-variables (standardized variables) and X-variables respectively
- Interpret the loadings for PC2 (easiest for Z-loadings)
- Calculate the PC2 scores for the 54 countries
- Using ***sort***, find the 10 "best" countries with respect to PC2(lowest scores)

- Make a scatter plot for the PC1 and PC2 scores
- Make a two-dimensional biplot for male time-data PC1 and PC2 scores and loadings
- Optionally, also make a three-dimensional biplot for male time-data PC1, PC2 and PC3 scores and loadings

- Calculate the sample correlations between PC1 and the original variables and likewise for PC2 and interpret the values
- Make a correlation circle diagram where the axes are the correlation between original variables and PC1 and PC2, respectively, and where each of the 8 disciplines then is a point in this coordinate system
- How close are the 8 points to the unit circle and what is the conclusion ?
- Calculate approximate simultaneous Bonferroni confidence intervals for the variances of time-data PC1 and PC2
- Conclude on the PCA

Convert the time dataset to a speed dataset:

- Calculate the speed in [m/sec] for all the 54*8 track records
- Calculate descriptive statistics S and R for the speed-data

Now, all data are commensurable and it is therefore desired to perform a renewed PCA, now using the sample covariance matrix, S, for the speed-data

PCA on S (for speed-data):

- Redo all calculations, plots and interpretations again, replacing the time-data with the speed-data and using S for the speed-data for the PCA
- Are there any major differences between the results and conclusions of the two ways of performing a PCA for the track records data ?

Problem 8.2 (Theoretical + MATLAB)

PCA for bivariate distribution

Consider the general population covariance and correlation matrices for a bivariate model

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

$$\mathbf{\rho} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

PCA on $\mathbf{\Sigma}$ for the bivariate model:

- Derive expressions for the two eigenvalues of $\mathbf{\Sigma}$
- Indicate the total variation of $\mathbf{\Sigma}$
- Indicate the fraction of the total variation explained by each of the two Principal Components (PC's)
- Derive expressions for the loadings for the two PC's (eigenvectors of $\mathbf{\Sigma}$)

PCA on $\mathbf{\rho}$ for the bivariate model:

- Derive expressions for the two eigenvalues of $\mathbf{\rho}$
- Indicate the total variation of $\mathbf{\rho}$
- Indicate the fraction of the total variation explained by each of the two PC's
- Derive expressions for the loadings for the two PC's (eigenvectors of $\mathbf{\rho}$)

Verification of results:

- Verify the derivations in MATLAB by comparing the results from above with the results achieved using command ***svd*** for numerical example values of the parameters (be sure to try both positive and negative values for ρ)

Problem 8.3 (Theoretical + MATLAB)

PCA calculation based on original variables (data matrices X or Z) instead of based on descriptive statistics (matrices S or R)

In this problem the design matrix (data matrix), X, is assumed to be of dimension (n x p) and without loss of generality the data are assumed to be centered (the sample mean have been subtracted from each of the p variables, thus, their sample mean are zero).

Alternative calculation of PCA(S):

Consider the usual PCA spectral decomposition of the sample covariance matrix

$$S \stackrel{\text{def}}{=} \frac{1}{n-1} X^T X = \sum_{i=1}^p \lambda_{S,i} e_{S,i} e_{S,i}^T = E_S \Lambda_S E_S^T$$

- Indicate the X-loadings, scores and variance for each of the p Principal Components (PC's)

Next consider the singular value decomposition (SVD) of a modified data matrix, Y:

$$Y \stackrel{\text{def}}{=} \frac{1}{\sqrt{n-1}} X = \sum_{i=1}^p \lambda_{Y,i} u_{Y,i} v_{Y,i}^T = U_Y D_Y P_Y^T$$

- Show that $Y^T Y = S$
- Show that $Y^T Y = P_Y D_Y^2 P_Y^T$
- Use this to conclude that a PCA on S can alternatively be done by calculating an SVD on Y and indicate the loadings, scores and variance for each of the p Principal Components (PC's), now expressed in the components of this SVD
- Verify your results using MATLAB

Alternative calculation of PCA(R):

Next, consider the usual PCA spectral decomposition of the sample correlation matrix

$$R \stackrel{\text{def}}{=} V^{-1/2} S V^{-1/2} = \sum_{i=1}^p \lambda_{R,i} e_{R,i} e_{R,i}^T = E_R \Lambda_R E_R^T$$

where V is the diagonal matrix containing the diagonal of S (the variances)

- Indicate the Z-loadings, X-loadings, scores and variance for each of the p Principal Components (PC's)

Next consider the singular value decomposition (SVD) of a modified data matrix, Z:

$$Z \stackrel{\text{def}}{=} \frac{1}{\sqrt{n-1}} X V^{-1/2} = \sum_{i=1}^p \lambda_{Z,i} u_{Z,i} v_{Z,i}^T = U_Z D_Z P_Z^T$$

- Show that $Z^T Z = R$
- Show that $Z^T Z = P_Z D_Z^2 P_Z^T$
- Use this to conclude that a PCA on R can alternatively be done by calculating an SVD on Z and indicate the Z-loadings, X-loadings, scores and variance for each of the p Principal Components (PC's), now expressed in the components of this SVD
- Verify your results using MATLAB